

GIS Analysis

Al Safa 2 Park — Professional GIS Layer Review & Data Gap Log

1. Status Overview

Professional GIS analysis (land use, building density, connectivity, slope, visibility, heat, noise, green coverage, service radius, walkability) normally requires authoritative datasets: Dubai Municipality GIS portal layers, satellite/aerial imagery, a DEM, and population/land-use shapefiles. **None of these raw datasets were included in the competition files provided** (only a JPG location graphic, a PDF brief, a DWG as-built, and a park design manual PDF). This document records what can legitimately be assessed now vs. what is a logged data gap.

2. Assessable Now (from provided materials)

Layer	Finding	Source
Land Use (qualitative)	Low-rise residential surrounds park on 3 sides; SZR + Al Qouz Industrial on 4th (east) side	Master-plan location image
Connectivity (qualitative)	Local street grid feeds the park; SZR is a hard edge/barrier on the east	Master-plan location image
Green Coverage (approx.)	~0.94% of the reviewed graphic's frame flagged as green-toned pixels via color-threshold script — a rough visual proxy only, not survey-grade canopy measurement	Python image analysis (Phase 1.02)
Walkability (qualitative)	Park is walkable from surrounding residential streets; crossing SZR to reach metro/Al Qouz side is not pedestrian-friendly	Contextual reading

3. Logged Data Gaps (require external GIS sourcing)

Layer	Requirement
Building Density	Dubai Municipality building footprint GIS layer
Slope	Site DEM / survey contours (may exist inside the DWG once converted)
Visibility analysis	Requires 3D massing model of surrounding buildings
Heat (urban heat island)	Requires satellite thermal imagery (e.g., Landsat/Sentinel thermal bands)
Noise	Requires traffic count data for SZR + acoustic modeling
Service radius / catchment	Requires population density GIS layer for accurate isochrone modeling

4. Recommendation

If genuine GIS rigor is required for the submission, the highest-value next step is: (1) convert the as-built DWG to DXF to recover any survey/contour layer; (2) optionally pull free satellite imagery for a heat/greenery cross-check — technically feasible with Python but requires external API access/credentials not yet confirmed.

This keeps the analysis honest: qualitative/contextual GIS reading is complete; quantitative GIS layers remain flagged as pending real data, not fabricated.